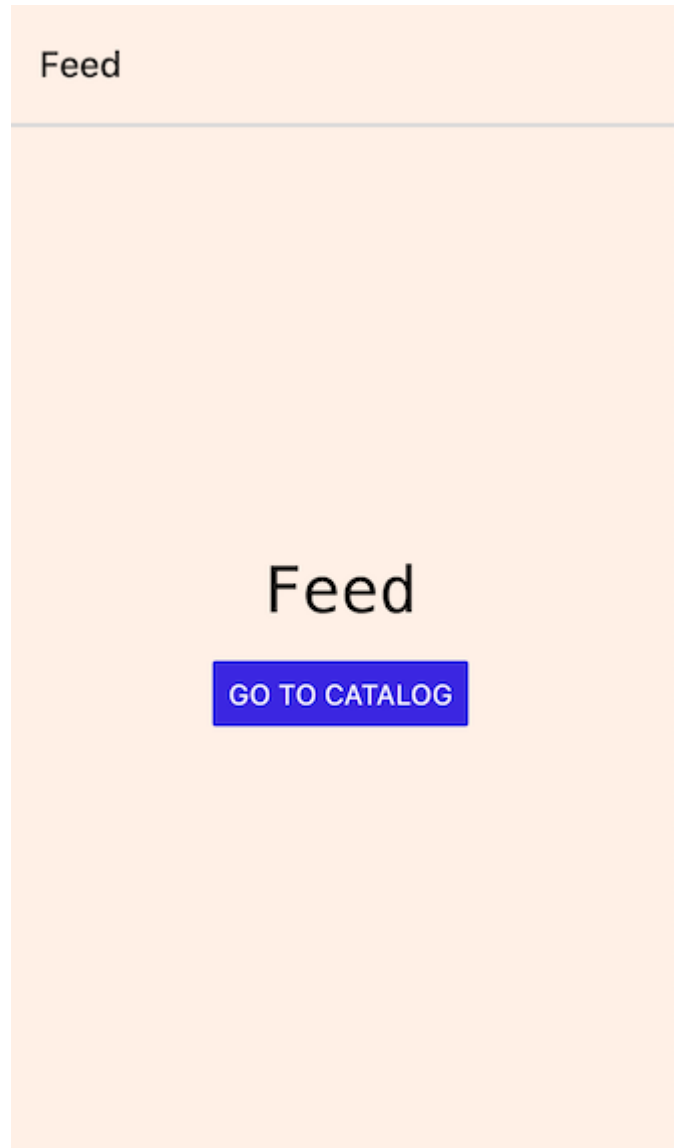


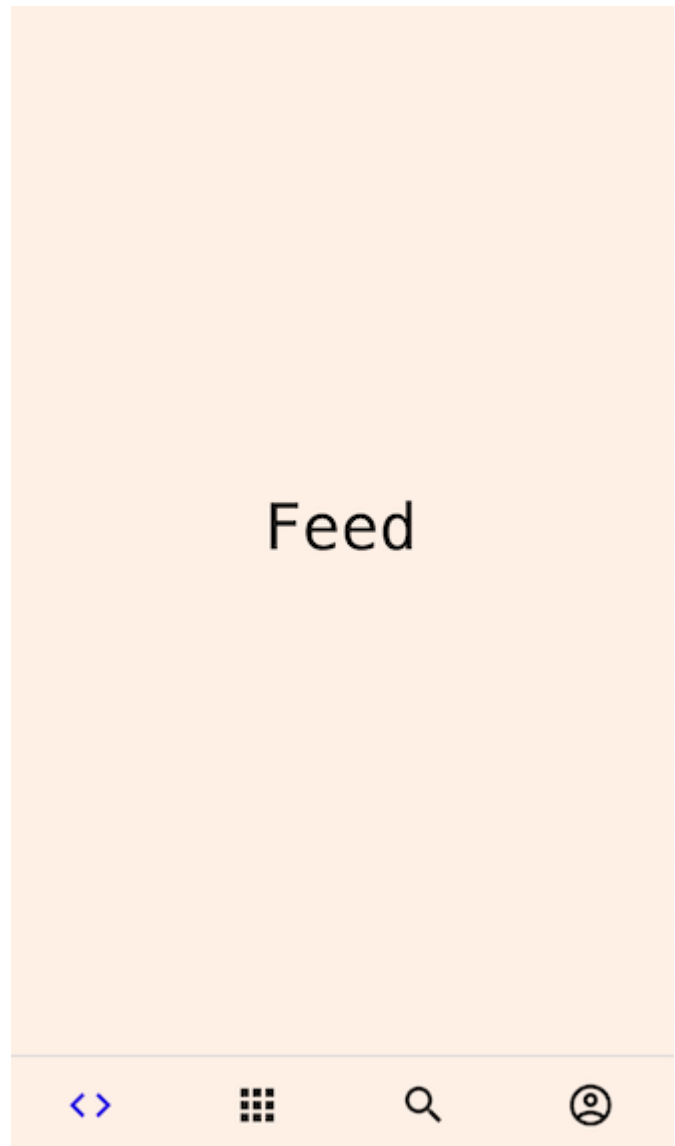
Stack Navigation

In the *stack navigation* pattern, a user has to go from screen to screen to navigate through all screens, where each one is pushed on a stack. The only UI rendered is a header with the screen title and a back button.



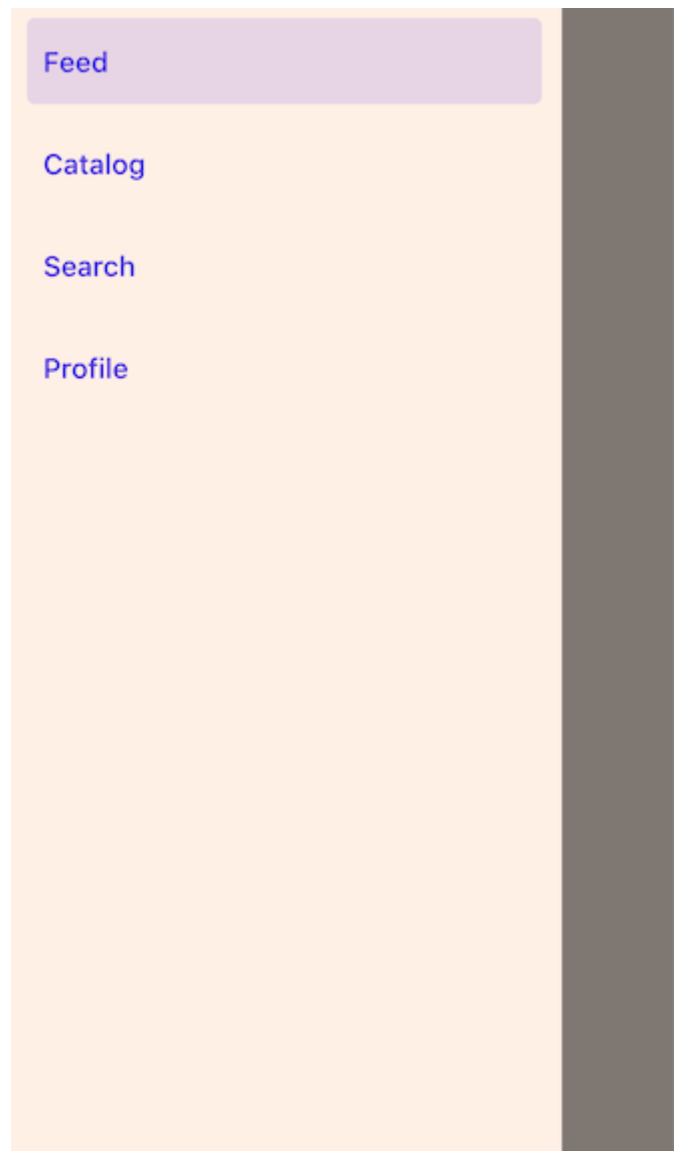
Tab Navigation

In the *tab navigation* pattern, a user uses a tab bar to switch between screens.



Drawer Navigation

In the *drawer navigation* pattern, a user uses a pane that can be opened by either swiping or tapping a button, which provides a menu where users can switch between screens.



NavigationContainer Component

In the `react-navigation` library, components to be organized must be wrapped in a `NavigationContainer` component since it keeps track of the navigation structure and makes sure the navigators can operate.

```
import { NavigationContainer } from
 '@react-navigation/native';

const App = () => (
  <NavigationContainer>
    { /* Insert your navigators and
content here */ }
  </NavigationContainer>
);
```

createStackNavigator Factory Method

In the `react-navigation` library, the stack navigator is created by the `createStackNavigator` factory method.

```
const Stack = createStackNavigator();

<Stack.Navigator>
  <Stack.Screen name="Feed" component=
{FeedScreen} />
  <Stack.Screen name="Catalog" component=
{CatalogScreen} />
</Stack.Navigator>
```

createBottomTabNavigator Factory Method

In the `react-navigation` library, the bottom tab navigator is created by the `createBottomTabNavigator` factory method.

```
const Tab = createBottomTabNavigator();

<Tab.Navigator>
  <Tab.Screen name="Feed" component=
{FeedScreen} />
  <Tab.Screen name="Catalog" component=
{CatalogScreen} />
</Tab.Navigator>
```

useNavigation Hook

In the `react-navigation` library, the `useNavigation` hook provides access to the navigation API and can be used to move users to different screens. It returns an object which is also passed as a `navigation` prop to screens and has multiple methods, including `navigate` (takes a screen name argument) and `goBack`.

// Using properties, only available in screen components

```
const FeedScreen = (props) => {  
  const nav = props.navigation;  
  
  return (  
    <Button  
      title="Go to home"  
      onPress={() =>  
nav.navigate('Home')}  
    />  
  );  
};
```

// Using the hook, available in all components

```
const HomeButton = () => {  
  const nav = useNavigation();  
  
  return (  
    <Button  
      title="Go to home"  
      onPress={() =>  
nav.navigate('Home')}  
    />  
  );  
};
```

Factory Methods

In the `react-navigation` library, all navigators are created by factory methods that use a naming pattern like `create<type>Navigator()`, which returns an object with `Navigator` and `Screen` properties. These properties are unique components that you must use when rendering the navigation structure.

```
const Stack = createStackNavigator();  
const Tab = createBottomTabNavigator();
```

// You can replace Tab with any other factory method result.

```
<Tab.Navigator>  
  <Tab.Screen name="Feed" component=  
    {FeedScreen} />  
  <Tab.Screen name="Catalog" component=  
    {CatalogScreen} />  
</Tab.Navigator>
```

